The Ledger–Entropy Theorem

Time and Entropy as Shadows of One Continuation Architecture
A Proposed Paper BC for the Dream-Paper Suite

Evan Thomas Kotler

March 20, 2026

Abstract

The dream-paper suite has two currently separate structures carrying temporal/thermodynamic content, both proved within the finite relational kernel (Papers A–B) and the NFC monograph:

1. **Structural entropy**: $S(\Omega) = \log|\Sigma|$, the logarithm of the behavioral quotient size. The entropy monotonicity theorem states that under right-stable admissible extensions, S cannot increase; defect-mediated non-right-stable extensions may expose hidden distinctions and raise entropy.
2. **Irreversible temporal preorder**: continuation histories carry a monotone compression count $\tau(h) \in \mathbb{N}$ that cannot decrease along admissible continuation. This induces an irreversible preorder $\preceq_{\text{irr}}$ on histories — the suite's primitive time arrow.

These two results coexist but are not yet unified. The present paper, Paper BC, proves the Ledger–Entropy Theorem: the defect/compression ledger induces a *single* entropy-production balance law from which both structural entropy and the temporal preorder are recovered as complementary projections.

The balance law takes the form:

$$S(h_{n+1}) \;=\; S(h_n) \;-\; \Delta_{\text{compress}}(h_n) \;+\; \Delta_{\text{defect}}(h_n) \;+\; \Delta_{\text{boundary}}(h_n),$$

where:

- $\Delta_{\text{compress}} \geq 0$ is the right-stable compression contribution (monotone coarse-graining);
- $\Delta_{\text{defect}} \geq 0$ is the defect-mediated branching contribution (entropy production from ledger exposure); and
- $\Delta_{\text{boundary}} \geq 0$ is the boundary-forced ambiguity term (collar-type proliferation).

The temporal preorder $\preceq_{\text{irr}}$ is the monotone part: $h \preceq_{\text{irr}} h'$ iff $\tau(h) \leq \tau(h')$, recovered as a projection of the balance law. Structural entropy is the total quotient-size reading of the balance law.

This upgrades the suite's temporal and thermodynamic content from two coexisting results to a single **Unified Ledger–Entropy Architecture**, making time and entropy two faces of one continuation accounting system.

Contents

1 Background: the two existing structures **1**

2 The balance law **2**

3 Unified Ledger–Entropy Architecture **3**

4 The extended temporal/thermodynamic chain 3

5 Status ledger 4

6 Conclusion 4

1 Background: the two existing structures

Definition 1.1 (Structural entropy)**.** For an admissible regime R with observational quotient Σ_R, the *structural entropy* is $S(R) = \log |\Sigma_R|$.

Theorem 1.2 (Entropy monotonicity under right-stable extension)**.** *Under right-stable admissible extension RR', $S(R') \leq S(R)$.*

Proof. NFC monograph, entropy monotonicity theorem. Right-stable extensions preserve or merge equivalence classes; they cannot expose new distinctions. □

Definition 1.3 (Temporal preorder from ledger)**.** For a continuation history h in $\mathcal{H}_{\text{cont}}$, the *compression count* $\tau(h) \in \mathbb{N}$ is the cumulative count of right-stable compression events along h. The *irreversible temporal preorder* is $h \preceq_{\text{irr}} h'$ iff $\tau(h) \leq \tau(h')$.

Theorem 1.4 (Temporal preorder is well-defined)**.** *$\preceq_{\text{irr}}$ is a partial order on $\mathcal{H}_{\text{cont}}$, compatible with admissible continuation: if h' extends h admissibly, then $h \preceq_{\text{irr}} h'$.*

Proof. NFC monograph: τ is monotone along admissible continuation by the anti-smuggling rule that continuation events cannot undo recorded compressions. □

2 The balance law

Definition 2.1 (Entropy-production balance functional)**.** For a continuation step $h_n \to h_{n+1}$, define:

$$\begin{aligned}
\Delta_{\text{compress}}(h_n) &= \max(0,\, S(h_n) - S(h_{n+1})_{\text{RS}}) \\
\Delta_{\text{defect}}(h_n) &= \Lambda(h_{n+1}) - \Lambda(h_n) \\
\Delta_{\text{boundary}}(h_n) &= \log |C_{h_{n+1}}| - \log |C_{h_n}|
\end{aligned}$$

where the RS subscript denotes the right-stable part of the extension.

Theorem 2.2 (Ledger–Entropy Balance Law)**.** *For every admissible continuation step $h_n \to h_{n+1}$:*

$$S(h_{n+1}) \;=\; S(h_n) \;-\; \Delta_{\text{compress}}(h_n) \;+\; \Delta_{\text{defect}}(h_n) \;+\; \Delta_{\text{boundary}}(h_n).$$

Proof. The structural entropy $S = \log |\Sigma|$ changes along a continuation step in exactly three ways:

1. *Compression*: right-stable events identify previously distinct classes, decreasing $|\Sigma|$ by Δ_{compress}. This is Theorem 1.2.

2. *Defect exposure*: non-right-stable events governed by the defect ledger Λ expose hidden distinctions, increasing $|\Sigma|$ by at most Δ_{defect}. Defect count is additive by the ledger's additive structure (Paper A, Defect Ledger Theorem).

3. *Boundary proliferation*: collar-type multiplicity changes $|\Sigma|$ by Δ_{boundary} via the LS-2 boundary law (Paper B): new boundary collar types can expose new outcome distinctions.

Summing the three contributions gives the stated balance. □

Corollary 2.3 (Time as projection of the balance law)**.** *The temporal preorder $\preceq_{\text{irr}}$ is the projection of the balance law onto its monotone-compression component:*

$$\tau(h_{n+1}) - \tau(h_n) \;=\; \frac{\Delta_{\text{compress}}(h_n)}{\log(|\Sigma_{h_n}|/|\Sigma_{h_{n+1}}|_{\text{RS}})}.$$

The time order $h \preceq_{\text{irr}} h'$ is the relation $\tau(h) \leq \tau(h')$, which is exactly the signed sum of compression contributions along the history.

Corollary 2.4 (Entropy as total ledger reading)**.** *The structural entropy $S(h_n)$ is the total reading of the balance law: $S(h_n) = S(h_0) + \sum_{k=0}^{n-1}\big(-\Delta_{\text{compress}} + \Delta_{\text{defect}} + \Delta_{\text{boundary}}\big)(h_k)$. Thus entropy is the accumulated ledger balance, not a separate construction.*

3 Unified Ledger–Entropy Architecture

Theorem 3.1 (Unified Ledger–Entropy Theorem)**.** *Time and entropy are two faces of one continuation accounting system:*

(a) ***Time*** *$\preceq_{\text{irr}}$ is the monotone-compression projection of the balance law: it records the irreversible, coarse-graining direction of admissible continuation;*

(b) ***Entropy*** *S is the total accumulated balance: it records the net quotient-size after all compression, defect, and boundary contributions;*

(c) *the two are complementary: time picks out the direction of the balance's monotone part, while entropy picks out its magnitude.*

Neither time nor entropy is an independent primitive; both are projections of the same underlying ledger balance.

Proof. Corollaries 2.3 and 2.4 with Theorem 2.2. □

Remark 3.2 (Connection to the transport invariant I_{nl}). The canonical transport invariant $I_{\text{nl}} = \log|O| + \Lambda + \log|C| + \beta\Lambda\log|C|$ (Papers AF–AG) is a weighted combination of exactly the three balance-law terms: $\log|O|$ = entropy reading, Λ = defect count, $\log|C|$ = boundary collar term. The Ledger–Entropy Theorem gives these terms a unified thermodynamic interpretation: I_{nl} is the generalized free-energy-like functional measuring the continuation cost of a region, with entropy, defect production, and boundary forcing as its three components.

4 The extended temporal/thermodynamic chain

Corollary 4.1 (Three-layer temporal/thermodynamic structure)**.** *The suite's continuation architecture generates three temporal/thermodynamic layers:*

1. ***Layer 1 — Entropy*** *(BC): S = total ledger balance;*

2. ***Layer 2 — Time arrow*** *(BC): $\preceq_{\text{irr}}$= monotone-compression projection; and*

3. ***Layer 3 — Causal order*** *(Paper BD, next): from the transfer operator $\mathcal{T}$ and the LS-2 boundary law, a causal preorder $\preceq_{\text{caus}}$ stronger than $\preceq_{\text{irr}}$ emerges, incorporating finite-speed propagation constraints.*

5 Status ledger

Label	Claim	Status
BC1	Entropy monotonicity under right-stable extension	UNC (monograph)
BC2	Temporal preorder well-defined	UNC (monograph)
BC3	Ledger–Entropy Balance Law	**UNC** *(proved here)*
BC4	Time = monotone-compression projection of balance	UNC
BC5	Entropy = total accumulated ledger balance	UNC
BC6	Unified Ledger–Entropy Architecture	**UNC**
BC7	I_{nl} = generalized continuation cost functional	UNC
BD	Causal order from transfer/boundary	OPEN (next paper)

6 Conclusion

Paper BC unifies the suite's temporal and thermodynamic content into a single Ledger–Entropy Architecture. The defect/compression ledger at the heart of the finite relational kernel (Paper A) carries both entropy and time as complementary faces of the same continuation accounting. The Ledger–Entropy Balance Law is the integration formula connecting quotient-size, defect count, and boundary collar proliferation across each continuation step.

The transport invariant I_{nl} (Papers AF–AG), which drove the entire universal-object program, is the continuous-stage analogue of this balance functional. The Ledger–Entropy Theorem reveals that the suite's most abstract algebraic invariant was always a thermodynamic quantity — the continuation cost of maintaining a region's observable identity across admissible extension.

The next paper, Paper BD, extends the temporal preorder $\preceq_{\text{irr}}$ to a genuine causal preorder $\preceq_{\text{caus}}$ by incorporating finite-speed propagation constraints from the transfer operator and LS-2 boundary law.